

Impact of design on manufacturing

All products carry two kinds of costs: the first is the one-time development cost called non-recurring expenses (NRE) and the second is the recurring costs (RE), which is actually the cost of the materials, manufacturing, etc. It may look that the impact of NRE is minimal, if you refer to Fig. 4, which shows the relationship between the cost of the changes in the design that need to be adopted into manufacturing and the maturity of the development cycle. It clearly shows that if the design changes have to be made in the later phase of product development, the cost impact will be very high. This highlights how important it is to address the manufacturing issues up front during the design phase itself, which will bring both NRE and RE down.

From the above example, it is clear that designing products, specifically keeping in mind the process of manufacturing (DFx), helps in controlling the cost. While there are multiple parameters that need to be taken into account during the design, as part of DFx, there are five key elements that contribute the maximum to the product cost and development time.

DFx: The big five

In design for x, 'x' stands for cost (DFC), assembly (DFA), manufacturing (DFM), reliability (DFR), testing (DFT), support (DFS), process (DFP), etc. Of these, the five that contribute the maximum are:

1. Design for cost
2. Design for assembly
3. Design for manufacturing
4. Design for test
5. Design for support

While looking at each of these aspects in detail, we will also look at the linkages between each of them so that we know exactly what should be done for effective implementation of these five parameters.

Design for cost (DFC)

Most designers feel that the cost of a product depends entirely on selecting

TABLE I Factors that Impact Designing for Cost	
Constituent factors of cost	Impact
Life of the selected part	A part may be cheaper at the time of designing. Yet, unless the life time of the product is ascertained, the part's availability may become expensive. This can delay the product launch. If the product doesn't meet the estimated volume, most vendors don't sustain it. Due to this, the design vendors take risk by incorporating conditions for the last time buy period in the supply contract.
Lead time for availability of the parts selected	More than the cost, sometimes even low-cost parts take 8-12 weeks from the date of purchase order. Often, designers take a decision using the sample available, as the actual availability may take more time, leading to either over stocking of parts or delay in shipments.
Wrong standardisation	Based on common sense, the designers tend to standardise the part, unmindful of the fact that their decision may lead to product failure. Sometimes, designers even interchangeably use carbon film resistors and metal film resistors (MFR) while MFRs are not recommended for high voltage applications.
Paneling of PCBs	One of the easiest ways of reducing product cost is to use paneling where a single large PCB contains multiple images of the PCB to be manufactured. This allows the assembly process and PCB fabrication costs to be reduced.

low-cost parts. This is a misconception. The cost of the part is, possibly, just one aspect, especially at the start of the design. Usually, by the time a product gets to the assembly line, the cost structure of the product changes due to various factors. Some of the factors that impact designing for cost are shown in Table I.

Design for assembly (DFA)

Design for assembly is a major area that can make or mar a product. Designers should therefore pay particular attention to this area. This area also has impact on the cost of the product and its support in the field. Design for assembly is necessary for designing of electronic and mechanical products, cable harness and software. Some aspects of designing for assembly and its impact is shown in Table II.

Design for manufacturing (DFM)

Most designers don't realise the importance of this key phase that leads to unexpected escalation in the cost of the product. Some of the important manufacturing aspects are listed in Table III along with their impact on the product.

Another important aspect of designing for manufacture is that it is interlinked with other DFx elements, leading to complex situations where any DFM solution needs to be validated with other DFx elements. For example, if by looking at cost cutting, one tries to use a low-cost part, this could possibly mean a lengthy manufacturing process, in turn, leading to cost escalation. So one of the key aspects of DFx is to look at the big picture in order to arrive at the optimal solution rather than looking at compartments.

Design for test (DFT)

Design for test is one of most neglected aspects by almost 95 per cent of the designers, leading to several yield problems as well as cost escalations. Typically, a product test strategy can affect the product in its quality/reliability as well as yield in the manufacturing. However, the strategy to test a product has to be decided much ahead in the design curve as it needs support from the design team in making the design testable, which involves both the design team and the test engineering team. Additionally, the design team needs to understand the basics of test engineering.